

Chilled Water Cooling Coil Detailed Engineering Report - v2.5

Prepared by user: sample review

Physical flow geometry: CROSS-FLOW (air perpendicular to tube/coolant direction). Water enters air-leaving side (cross-counterflow tendency)

Design Inputs

Input / construction	Value
Face W x H	1.100 x 0.559 m
Number of tube rows (airflow depth)	4
Nominal tube-bank depth	88.0 mm
Fin type	Wavy fin
Tube cross-section / fin construction	Flat / elliptical tube / Serpentine fins
Fin material	Aluminum
Tube material	Copper
FPI / fin pitch	12.0 1/in / 2.117 mm
Fin thickness	0.120 mm
Tube major / minor OD / wall	9.530 / 6.000 / 0.350 mm
Pt / Pl	25.400 / 22.000 mm
Wave parameter Pd / half-period xf	1.000 / 1.000 mm
Air HTC / dry dP calibration	1.000 / 1.000
Wet/dry air dP ratio	1.120
Air / water fouling resistance	0.000000 / 0.000000 m ² .K/W
Airflow	1.1000 m ³ /s = 3960 m ³ /h = 2330 CFM
Gross face velocity	1.790 m/s
Entering air DB / RH	27.00 degC / 50.0%
Coolant / concentration	Water / 0.0%
Coolant inlet / pressure	7.00 degC / 300.0 kPa abs
Coolant mass flow / circuits	1.2000 kg/s / 11
Coolant volume flow	4.3206 m ³ /h = 1.2002 L/s
Coolant flow input unit	Volume flow (m ³ /h)
Suggested start tube velocity	1.000 m/s
Suggested start coolant flow	1.4555 m ³ /h = 0.4043 L/s = 0.4043 kg/s
Approx tube velocity from entered flow	2.968 m/s
Circuit tube-end arrangement	Same tube end (even passes/circuit required)
Calculated tubes / row / total	22 / 88
Circuit model	Equal-flow circuit-count model
Circuit pass counts	Not explicitly routed
Supply header OD / wall	42.40 / 1.50 mm
Return header OD / wall	42.40 / 1.50 mm
Header length / arrangement	0.559 m / Opposite-end (reverse-return tendency)
Header feed end	Top
Return bend / branch K	1.500 / 0.500
Common inlet / outlet K	0.500 / 1.000
Tube / header roughness	1.50 / 1.50 um
Water row progression	Water enters air-leaving side (cross-counterflow tendency)

Performance

Performance	Value
Total capacity	23.10 kW
Sensible / latent	15.16 / 7.94 kW
SHR	0.656
Leaving air DB / WB	15.37 / 13.33 degC
Leaving air RH	80.0%
Leaving coolant	11.59 degC
Condensate	11.26 kg/h
Wet fraction / state	100.0% / Fully wet by row
Air dP	18.6 Pa
Water dP avg	218.85 kPa
Water dP min / max	218.83 / 218.86 kPa

Psychrometric & Energy Balance

Psychrometric / energy state	Value
Entering air DB / WB / RH	27.00 / 19.53 degC / 50.00%
Entering humidity ratio	0.011196 kg/kg dry air
Entering air enthalpy	55.711 kJ/kg dry air
Entering dew point	15.70 degC
Leaving air DB / WB / RH	15.37 / 13.33 degC / 80.01%
Leaving humidity ratio	0.008734 kg/kg dry air
Leaving air enthalpy	37.541 kJ/kg dry air
Dry-air mass flow	1.2713 kg/s
Coolant density / cp	999.86 kg/m3 / 4.1956 kJ/kg.K
Coolant viscosity / conductivity	0.00133269 Pa.s / 0.5774 W/m.K
Q from air enthalpy balance	23.099 kW
Q from coolant balance	23.099 kW
Independent air-water balance error	0.0012%

Heat Transfer Diagnostics

Heat-transfer diagnostic	Value
C air	1.305 kW/K
C coolant	5.035 kW/K
Cmin / Cmax	1.305 / 5.035 kW/K
Capacity ratio Cr	0.259
Dry-reference NTU	1.159
Dry cross-flow effectiveness	0.629
Wet enthalpy effectiveness	0.551
Air temperature effectiveness	0.582
Air Re / Pr	1181 / 0.713
Air-side h	44.79 W/m ² .K
Air-side Colburn j / friction f	0.012160 / 0.052684
Segmented dry air dP before wet factor	16.99 Pa
Final wet-coil air dP	18.57 Pa
Coolant Re / Pr	14525 / 9.683
Water-side h	11214.29 W/m ² .K
Water Darcy friction factor	0.028528
Overall dry-reference UA	1512.66 W/K
Capacity-rate limiting side	Air side
Resistance limiting side	Air side
Resistance split air / water / wall	93.5% / 6.4% / 0.1%

Velocities

Velocity / flow item	Value
Gross air face velocity	1.790 m/s
Max air velocity between fins/tubes	2.478 m/s
Coolant velocity in one tube (avg)	2.968 m/s
Coolant tube velocity min / max	2.968 / 2.968 m/s
Supply header velocity	0.984 m/s
Return header velocity	0.984 m/s
Coolant mass flow per circuit	0.1091 kg/s

Hydraulic Pressure-Drop Breakdown

Hydraulic detail	Value
Supply header ID	39.40 mm
Return header ID	39.40 mm
Average total circuit path dP	218.849 kPa
Minimum / maximum path dP	218.828 / 218.863 kPa
Path spread	0.0344 kPa
Maximum flow maldistribution	0.000%
Hydraulic model	Equal-flow/header path model
Common inlet + outlet fitting dP avg	218.849 kPa
Component sum check	218.849 kPa

Calculated Geometry

Calculated geometry	Value
Tube rows in airflow direction	4
Tube and fin orientation	Tubes run along coil face width; major ellipse axis points into airflow depth. One serpentine strip f
Nominal tube-bank depth	88.0 mm
Tubes / row	22
Total tubes	88
Selected circuits	11
Circuit tube-end arrangement	Same tube end (even passes/circuit required)
Circuit model	Equal-flow circuit-count model
Tube length	1.100 m
Fin count	520
Fin construction / serpentine strips	Serpentine fins / 21
Fin area per plate / individual strip	0.00343 m2 (two sides)
Tube hole count / removed area per plate	0 / 0.00000 m2
Gross tube outside / covered by fins	2.3920 / 0.0000 m2
Tube inner flow area / hydraulic diameter	0.0000368 m2 / 6.522 mm
Face / free-flow area	0.615 / 0.444 m2
Free-area ratio	0.722
External air-side area (total)	39.90 m2
Fin area (net, two-sided)	37.51 m2
Exposed tube outside area	2.39 m2
Inside tube heat-transfer area	2.18 m2
Air-side correlation	Wavy fin - Wang plain-fin baseline with corrected physical fin/tube area

Engineering Warnings

- Wavy-fin mode currently uses a transparent plain-fin Wang baseline with corrected physical fin/tube area. Calibrate h and dP against the actual wavy fin die before production use.
- Flat tubes or serpentine fins use round-tube air-side j/f correlations with a geometry-equivalent diameter. Flat-tube water HTC and pressure loss use hydraulic-diameter extensions of smooth circular-tube methods. Treat both capacity and pressure loss as screening estimates; calibrate against a matching core before manufacturing.
- High tube water velocity (>2.4 m/s): check erosion, noise and return-bend/header losses for your tube material and water quality.

Row-by-Row Thermal Marching

Rows are numbered in the air-flow direction. Physical geometry is CROSS-FLOW at every row. Water enters air-leaving side (cross-counterflow tendency). Converged: True in 26 iteration(s).

Row	Air DB in	Air DB out	Air WB out	RH out %	Water in	Water out	Q kW	Latent kW	Wet %	Mode
1	27.00	23.36	17.88	58.9	10.26	11.59	6.67	1.91	100.0	Fully wet
2	23.36	20.27	16.29	67.0	9.06	10.26	6.06	2.03	100.0	Fully wet
3	20.27	17.63	14.77	74.0	7.97	9.06	5.47	2.03	100.0	Fully wet
4	17.63	15.37	13.33	80.0	7.00	7.97	4.91	1.96	100.0	Fully wet

Method / Validation Note

- The physical fin-and-tube coil is cross-flow: air moves through the depth and coolant moves along the tubes, approximately 90 degrees to the air velocity.
- The water-row-progression option only changes which coil-depth side receives cold water. It is not a physical parallel-flow/counterflow change of the local air/tube orientation.
- Plain fins use the Wang, Chi & Chang (2000) plain fin j/f correlation. Wavy+louvered fins use the Wang-Tsai-Lu correlation as documented by ACHP.
- Wavy-only mode is deliberately labelled a calibration-required baseline rather than inventing unverified correlation coefficients.
- Water-side heat transfer uses Gnielinski with transition treatment; tube/header pressure loss uses Darcy-Weisbach with Churchill friction.
- With a complete circuit map, the app uses a fully coupled tube-by-tube / air-lane model. Each tube cell receives its own entering coolant temperature and resolved circuit flow and feeds its outlet states to the next coolant pass and next air row.
- Flat-tube air-side j/f and serpentine-fin j/f use an equivalent round-tube baseline: geometry, fin count, surface area, free flow and internal hydraulic diameter are calculated from the selected cross section, but capacity and air pressure loss require calibration against the actual core.
- Unequal circuit pass counts are allowed when each circuit retains the required same-end/even or opposite-end/odd outlet parity. The explicit hydraulic solver calculates the resulting flow maldistribution rather than assuming equal flow.
- The 2-D model assumes equal entering dry-air mass flow among vertical lanes and does not yet include lateral cross-fin conduction between adjacent tubes; these are explicit higher-order refinements.
- Validate air HTC, wet air dP and fitting K values against actual coil test or trusted manufacturer data before production release.